

Predicting Formal Protest Petitions Against Housing Development

Evidence from Austin, Texas

Daniel Hardesty Lewis

M.S. Candidate, Urban Planning

Graduate School of Architecture, Planning and Preservation

Columbia University

April 16, 2026

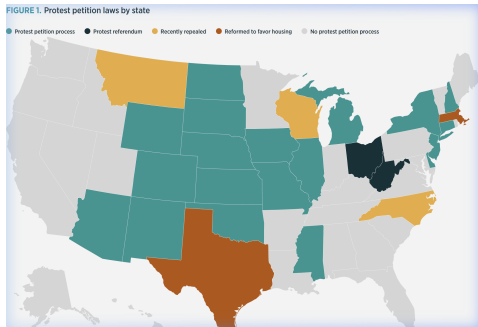
Housing Crisis and Local Opposition

The Affordability Problem

- Austin grew $\approx 40\%$ between 2010–2024.
- New housing requires discretionary rezoning approval.
- Projects face public hearings and organized opposition.

The Political Mechanism

Texas law gave adjacent property owners a formal institutional veto: the **protest petition**.

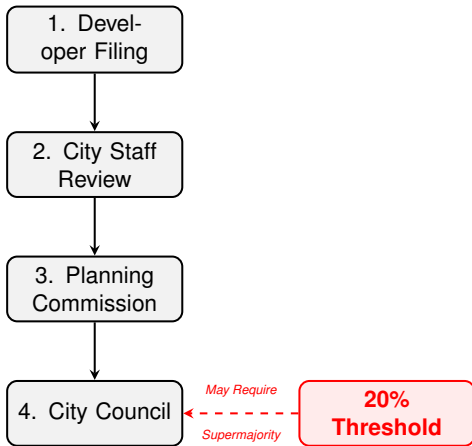


The Protest Petition: An Institutional Veto

Texas LGC Chapter 211: The 20% Rule

- Owners of 20% of land within 200ft may file written protest.
- A valid petition triggers a supermajority ($\frac{3}{4}$) council vote.
- Post-2014 council (11 members): 6 votes → 9 required.
- 21 states maintain similar petition statutes.

A localized minority can block citywide housing decisions.

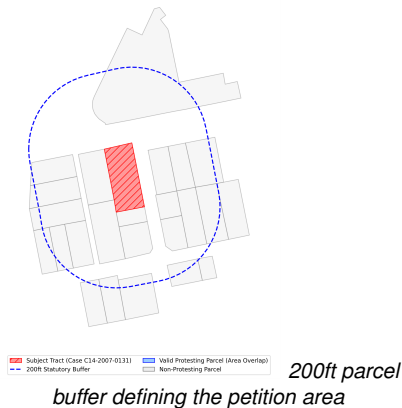


Protest Petition Rules Across States

State	Threshold	Buffer	Supermajority
Texas	20%	200ft	3/4
Iowa	20%	200ft	3/4
New York	20%	100ft	3/4
Colorado	20%	100ft	2/3
Oklahoma	50%	300ft	3/5 or 3/4
Massachusetts	50%	300ft	2/3

Source: Furth & McKinley (2025). 21 states maintain active protest petition statutes.

Visualizing the 200ft Statutory Protest Buffer
(Empirical TCAD Parcel Boundaries for Case C14-2007-0131)



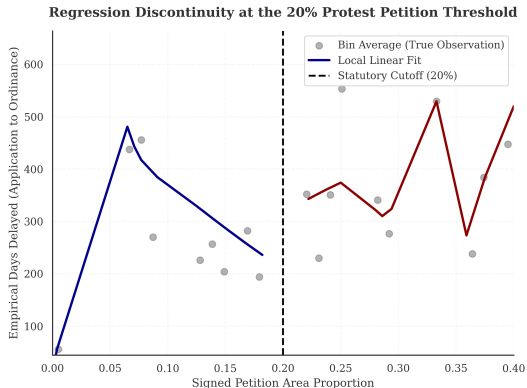
Why Petitions Matter: Institutional Costs

Reduced-Form Discontinuity at the 20% Threshold

- Running variable: reconstructed petition-share.
- Outcome: council processing time.
- Estimator: triangular-kernel WLS at the 20% margin.

Result: Crossing the threshold adds **42.5 days** (6.1 weeks).

95% CI: [26.4, 58.6] days. McCrary $p > 0.05$.



The Research Question

The Prediction Problem

Can we anticipate protest petitions *before* public hearings using only filing-date records?

Filing-Date Features

- Site geometry: lot area, height/FAR deltas.
- Current vs. requested zoning class.
- Buffer demographics and TCAD appraisals.
- Historical petition rates in the district.

A Sample Case

Feature	Case C14-2018-0156
Gross Site Area	4.92 Acres
Height Request	+15.5 ft
FAR Request	+0.43
Neighborhood	Riverside
Result	Valid Petition

Race/ethnicity excluded from predictive features; reserved for post-hoc equity audits only.

The Rarity Problem

Sample Characteristics

- $N = 6783$ cases, 2007–2024.
- Base rate: $\approx 2.6\%$ (1 in 38 cases).
- Test-set positives: $n_{\text{pos}} = 15$.

With so few events, confidence intervals are wide. The model output is a **score per case** (0–1); cases are ranked by that score, not classified by threshold.

Evaluation Design

- Train on years $\leq t$; evaluate on years $> t$.
- Temporal holdout for the test set; random cross-validation within training.

Primary metric: PR-AUC

AUROC omitted: we are predicting a rare class.

How We Arrived at N = 6783

Selection Step	Cases
Raw Austin open-data records	15,238
Discretionary cases only	7,153
Usable filing-year information	6783
Analytic sample	6783

Excluded: non-discretionary variances, ETJ cases (petition statute does not apply), and cases with unrecoverable timestamps.

Illustrative Panel Row

Case	Year	Lag Pet.	Med. Inc.	Own. Sh.
C14-2015-001	2015	3	\$85k	62%
C14-2015-002	2015	0	\$42k	21%
C14-2015-006	2015	7	\$112k	85%
C14-2016-009	2016	1	\$55k	34%
C14-2016-012	2016	4	\$95k	71%

Each row is one case; all features are available at filing date.

Timeline Integrity: As-Of Features

Enforcing the Constraint

- 17 years of zoning records from Austin's open-data portal (2007–2024).
- Filing dates back-calculated from statutory minimums.
- American Community Survey (ACS) and Travis County Appraisal District (TCAD) records anchored to publication date.
- $N = 8$ IRB-approved interviews (ACY0820) for institutional grounding.

COLUMBIA UNIVERSITY
Graduate School of Architecture, Planning and Preservation

Urban Studies and
City, Community, and Planning

Research Project Overview

Predicting NIMBYism in Austin's Housing Reform Era

Project Abstract

This research project explores the development of AI and its impact on the landscape of urban planning in Austin, Texas. By analyzing public data, including zoning records, public hearings, and community data, the study aims to understand the implications of using predictive models in urban planning. The project explores predictive modeling, natural language processing, and qualitative analysis to predict urban behavior, identify emerging patterns in opposition, and consider their implications for planning outcomes.

Study Objectives

Identify Patterns: Understand the spatial and temporal patterns in opposition to housing reform and identify demographic predictors of this opposition.

Evaluate Predictive Tools: Assess whether administrative data and predictive models can accurately forecast opposition in specific development projects.

Demographic Implications: Explore how demographic factors influence the likelihood of using administrative data in housing policy and identifying the ethical implications of using administrative data.

Participant Role

We are seeking perspectives from urban planners, developers, neighborhood advocates, and community organizers.

Availability: A brief initial interview is required, followed by a longer interview.

Topic: Your experience with housing, development planning tools, and community engagement in Austin.

Data Privacy and Ethical Standards

This study has been approved by the Columbia University Institutional Review Board under Protocol Y01304.

Participation is entirely voluntary.

You may decline to answer any specific question or discontinue the interview at any time.

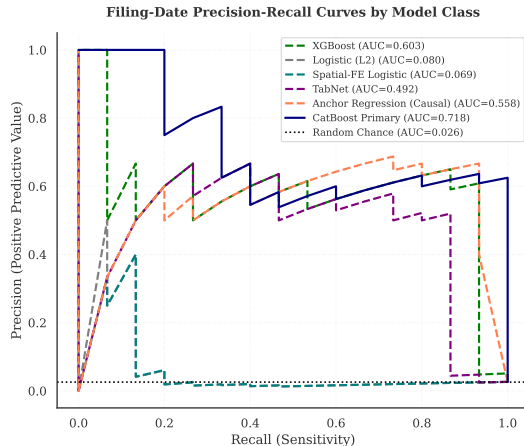
Identification with a key identifier will be removed from all resulting publications unless you grant explicit permission to be named.

Researcher Contact:
David H. Lee
Graduate School of
Architecture, Planning and
Preservation

IRB Contact:
Columbia University Mailstop 6100
Phone: 212-850-7440
Email: AUIR0820@columbia.edu

Predicting Resistance: Filing-Date Results

Tree Ensembles Dominate at the Top of the Risk Ranking



Operational Utility: Triage in Practice

Risk Ranking Performance

- Highest-risk 10% of cases: **25.4%** precision.
- **9.9x** lift over the base rate.
- Captures 15 true positives in 59 flagged cases.

Reviewing flagged cases is **nearly 10x** more likely to surface a genuine petition case.

Appropriate Uses

- ✓ Prioritize outreach for high-risk cases.
- ✓ Flag cases for early negotiation.
- × Hard probability cutoffs.

Four-Family Modeling Benchmark

Architecture Families Benchmarked

Family	Key Property
Tree (CatBoost)	Best tabular OOD PR-AUC
Deep (TabNet/MLP)	Requires L2; overfits at low N
Linear (L2 LR)	Strong at T+0; decays over time
Robust (V-REx)	Distributional robustness baseline

Selection Criterion

All families tuned to maximize out-of-distribution PR-AUC under strict temporal separation.

CatBoost remains the manuscript's benchmark tree model; this slide compares families on temporal PR-AUC, not calibration.

Reading the Attribution Charts: Feature Clusters

Cluster	Representative Variables
Parcel Scale	Lot acres, gross site area, deed acreage, lot size (sqft)
Zoning Density	Requested FAR delta, height delta, building coverage delta
Neighborhood Income	Median household income, gross rent, poverty count
Demographics	ACS % White, % Hispanic, % Black, % Asian
Housing Tenure	ACS owner-occupied, renter-occupied, total units
Property Valuation	TCAD appraised value, market value, land value
Historical Protest Activity	Prior protest flag, 1-year and 3-year spatial contagion

Variables with Spearman $|r| > 0.70$ are aggregated into clusters after model fitting; SHAP values are summed within clusters.

What Drives Petition Risk

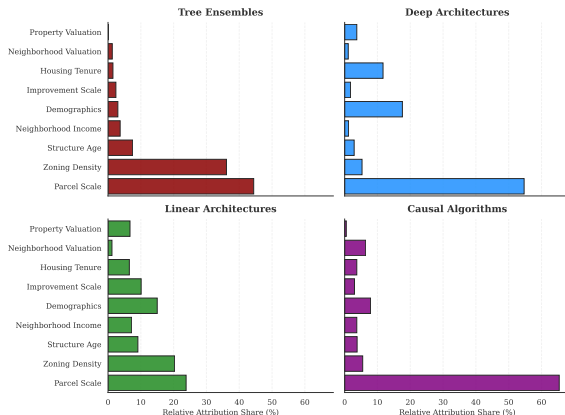
Key Finding

Tree ensembles weight the **physical ask**: parcel scale (44%) and zoning density request (36%) together account for 80%+ of attribution.

Deep models also weight parcel scale (55%) but add **neighborhood social fabric**: demographics (18%) and housing tenure (12%).

Same data, same outcome;
different internal logic.

Comparative Primary Reliance: Top Feature Clusters Across Architectures

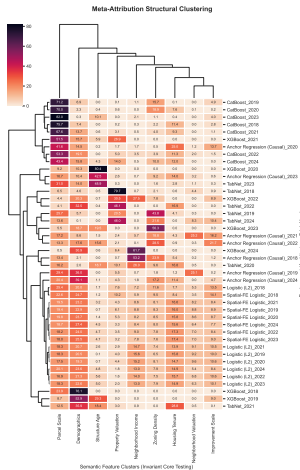


Architecture Determines Feature Reliance

Architecture-Dependent Reliance

- Trees: weight parcel scale and zoning-density request.
- Deep models: add neighborhood composition and tenure.
- Same data, same task: different internal logic.

Trees track the ask; deep models fold in who lives nearby. Those may reflect different opposition mechanisms.



Temporal Drift and Neighborhood Agency

When Does the Model Break?

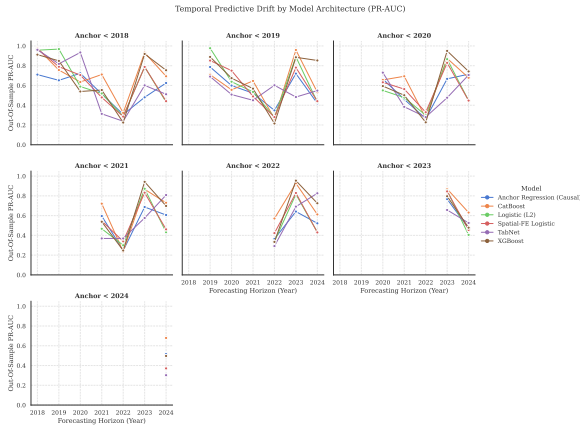
- Linear models collapse at T+3 horizons.
- Tree ensembles degrade more gracefully.

The 2022 Dip

- PR-AUC drops sharply across all families in 2022.
- Consistent with pandemic-era filing disruptions reaching the evaluation window.

Performance sensitivity to ambient shocks is *consistent with* mobilization as a strategic social choice.

Note: descriptive out-of-sample variation; no single institutional break is identified causally.

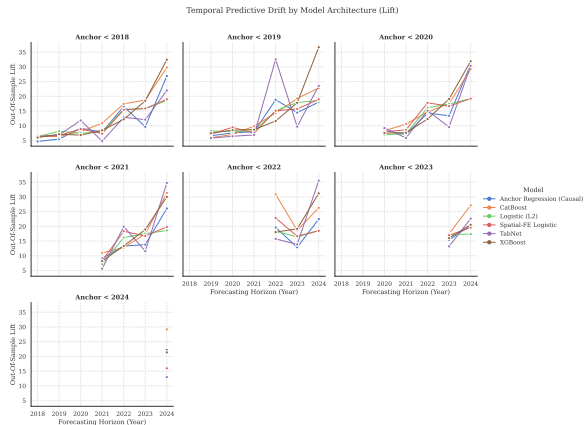


Lift Concentrates as the Regime Winds Down

A Counter-Intuitive Finding

- Lift rises sharply in 2022–2024 across all anchors and architectures.
- Models trained in any period achieve 20–40 $\times$ lift in 2024.

As the institutional regime declines, the cases that still attract petitions become more concentrated in the highest-risk situations. The model's risk rankings sharpen.



What the Algorithm Cannot See

What the Algorithm Cannot Distinguish

- Exclusionary NIMBYism (affluent West Austin blocking density).
- Anti-displacement resistance (East Austin protecting tenants).

Same risk weight; opposite social meaning.

Practitioner Perspectives ($N = 8$)

- *“Haters are overrepresented and silent supporters are underrepresented.”* (Beaudet) helps explain the performance ceiling.
- Torrado warns that rigid cutoffs can miss real displacement risk; use ranked triage, not hard thresholds.
- Tomko notes that formal opposition lacks anonymous channels, so the labels understate full community sentiment.

COLUMBIA UNIVERSITY Graduate School of Architecture, Planning and Preservation	
Predicting NIMBYism in Austin's Housing Reform Era	
Purpose	
To explore how a risk-weighted algorithmic approach to housing reform in Austin and how the perceived role of public policy in planning and zoning.	
Opening	
1. How do participants in the study (NIMBY) think they are the most likely to be affected by the study?	
2. How do participants in the study (NIMBY) think they are the most likely to be affected by the study?	
3. How do participants in the study (NIMBY) think they are the most likely to be affected by the study?	
Part 2: Background and Role	
1. How do participants in the study (NIMBY) think they are the most likely to be affected by the study?	
2. How do participants in the study (NIMBY) think they are the most likely to be affected by the study?	
3. How do participants in the study (NIMBY) think they are the most likely to be affected by the study?	
Part 3: Experiences with Neighborhood Opposition	
1. What are the most common experiences with neighborhood opposition in the study?	
2. How do participants in the study (NIMBY) think they are the most likely to be affected by the study?	
3. How do participants in the study (NIMBY) think they are the most likely to be affected by the study?	
4. How do participants in the study (NIMBY) think they are the most likely to be affected by the study?	
Part 4: Perceptions of Fairness and Participation	
1. How do participants in the study (NIMBY) think they are the most likely to be affected by the study?	
2. How do participants in the study (NIMBY) think they are the most likely to be affected by the study?	
3. How do participants in the study (NIMBY) think they are the most likely to be affected by the study?	
Part 5: Views on Predictive and Algorithmic Tools	
1. How do participants in the study (NIMBY) think they are the most likely to be affected by the study?	
2. How do participants in the study (NIMBY) think they are the most likely to be affected by the study?	
3. How do participants in the study (NIMBY) think they are the most likely to be affected by the study?	
4. How do participants in the study (NIMBY) think they are the most likely to be affected by the study?	
5. How do participants in the study (NIMBY) think they are the most likely to be affected by the study?	

Interview guide used under IRB protocol
ACY0820

Limitations

Methodological

- **Chilling effects:** Only filed cases observed; pre-emptive withdrawals are invisible. Selection correction was not feasible.
- **Timestamp imputation:** Milestone dates approximated from statutory minimums.
- **Unobserved certification:** Outcome is reconstructed area share, not the clerk's legal determination.

Ethical and Governance

- **Proxy discrimination:** Neighborhood income and demographics encode class and race dynamics even without protected-class inputs.
- **Motive blindness:** Cannot separate exclusionary NIMBYism from anti-displacement resistance.
- **External validity:** Parameters are Austin-specific and pre-HB 24 only.

Conclusion

What we showed

- Crossing 20% adds 42.5 days of processing time: the outcome is worth predicting.
- Filing-date data ranks petition risk above chance (PR-AUC = 0.718, CI: [0.48, 0.91]), with top-decile lift of 9.9x.
- Algorithm choice determines which feature clusters drive predictions, a normative decision.

What surprised us

- Retraining on modern data degrades lift; legacy models preserve spatial sorting intelligence.
- Performance ceilings are consistent with mobilization as a strategic social choice.

What it implies

- Predictions belong in relative triage rankings, not case-level probability thresholds.
- Methodology is portable; Austin-specific parameters are not.

Implications: A Now-Closed Policy Window

The HB 24 Shift (September 1, 2025)

- Texas HB 24 exempted residential upzonings from the supermajority requirement.
- The 20% institutional veto no longer applies to the most contested case types.
- The institutional regime is closed; its demographic and spatial correlates of opposition remain visible in the predictions.

What travels forward

- Methodology (time-stamped features, temporal holdouts, semantic clustering, qualitative integration) is portable to other discretionary review systems.
- New York, Iowa, Oklahoma, and Arizona retain active petition statutes. The framework applies wherever institutional veto mechanisms create predictable opposition patterns.

Discussion / Q&A

Thank you for your time and feedback.

Dory Kornfeld
Weiping Wu
James Piacentini
Josh Begley

Stijn Van Nieuwerburgh
Matthew Bauer
Adam Vosburgh

Selected References I

- Furth, S., & McKinley, K. (2025). *Protest petition statutes across U.S. states*. Mercatus Center.
- Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A.V., & Gulin, A. (2018). CatBoost: Unbiased boosting with categorical features. *NeurIPS*.
- Arik, S.O., & Pfister, T. (2021). TabNet: Attentive interpretable tabular learning. *AAAI*.
- Krueger, D., Caballero, E., Jacobsen, J.H., Zhang, A., Binas, J., Zhang, D., Le Priol, R., & Courville, A. (2021). Out-of-distribution generalization via risk extrapolation (V-REx). *ICML*.
- Lundberg, S.M., & Lee, S.I. (2017). A unified approach to interpreting model predictions. *NeurIPS*.
- Imbens, G., & Lemieux, T. (2008). Regression discontinuity designs: A guide to practice. *Journal of Econometrics*.
- Texas Local Government Code §211.006 (protest petition statute).
- Texas HB 24 (2025, 89th Legislature). An Act Relating to Zoning Authority.

Appendix: Hyperparameter Robustness

